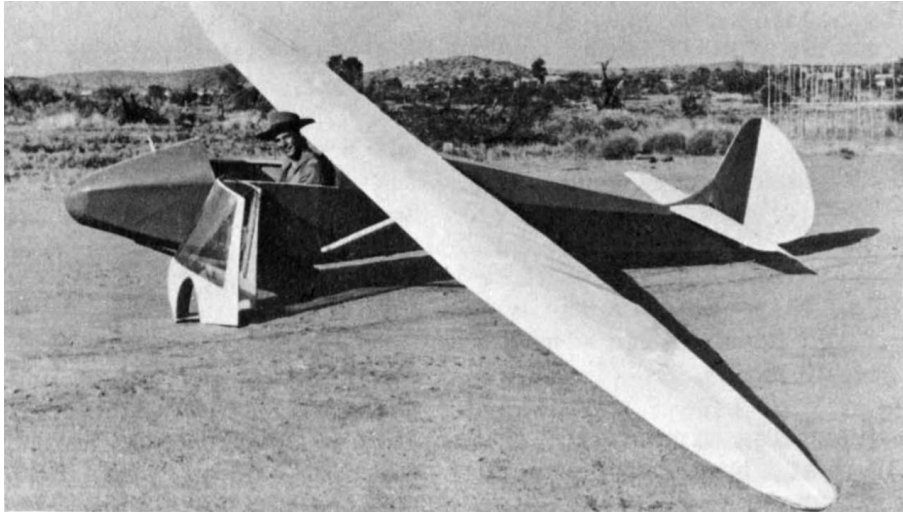


ACCIDENT REPORT

ACCIDENT TO J-2 "JUMBUCK" GLIDER VH-GLH,
NEAR ALICE SPRINGS, NORTHERN TERRITORY
on 24th NOVEMBER, 1963

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1. At 1615 hours C.S.T. on 24th November, 1963 a J2 "Jumbuck" glider, registered VH-GLH, struck the ground 1 mile north-west of the Alice Springs townsite aerodrome, Northern Territory, following loss of a major portion of the starboard mainplane,
2. The registered owner of the glider was Thomas Alfred Henry Bird of Alice Springs. The accident occurred during a landing approach following a private soaring flight of approximately one hour's duration.
3. The pilot, Miss Mary Elizabeth Crichton Weaving, was the sole occupant of the glider and she was killed in the accident.
4. The glider was substantially damaged by impact forces.
5. The pilot, who was 43 years of age, held A, B, C and Silver C pilot certificates issued by the Gliding Federation of Australia. She had completed 620 flights, involving a flight time of 121 hours, in gliders, of which 57 flights were in the J.2 glider.
6. The last Certificate of Airworthiness in respect of this glider expired on 13th September, 1962 and it was operated on a number of flights subsequent to that date, including the flight which culminated in this accident, without compliance with the provisions of Air Navigation Regulation 108(1)(b).
7. Inspection of the glider after the accident revealed that the tensile strength of all four aileron cables had been reduced to a low value as a result of extensive fretting, extreme brinelling and advanced fatigue and that the port upper aileron cable had failed as a result of an in-flight tensile load in excess of its residual strength.
8. The failure of this aileron cable permitted aileron rotation which coupled with mainplane bending to produce a flutter oscillation in the port wing and, by resonance, in the starboard wing. The very high bending loads produced by the flutter condition and aerodynamic loads arising from a substantial increase in wing incidence, as the wing structure progressively failed, probably combined to induce the failures found in both booms of the starboard main spar.
9. Although the failures in each of the spar booms occurred at points where the wood grain slopes were outside the normally accepted standards for aircraft construction it is considered that the ultimate wing strength of this glider was adequate for normal or aerobatic flight.

10. The last known inspection of the flying control system of this glider was carried out on 26th August, 1962 and the inspecting engineer then considered it to be airworthy. It is probable that the subsequent inspections required to be conducted for the re-validations of a Certificate of Airworthiness would have revealed the deteriorated state of the aileron cables.

11. There is no evidence to indicate that the glider was loaded other than within safe limits.

12. At the time of the accident there was no significant cloud, the wind was from 110 degrees at 7 knots, the surface temperature was 100 degrees Fahrenheit. The existence of dust devils in the area suggested the presence of thermal activity with some turbulence,

13. During the circuit preparatory to landing, and at a height of approximately 500 feet, observers on the ground noted the wing tips on VH-GLH vibrating violently. These vibrations were observed to occur for short periods on several occasions and were accompanied by a loud vibratory noise. The glider then stalled and commenced a spin from which recovery was effected. Further vibrations then occurred and another spin developed during which the outboard 10 feet of the starboard mainplane became separated and the glider spiralled to the ground.

14. CAUSE :

The cause of the accident was that the aileron control system of this glider had not been inspected with sufficient frequency to detect the serious deterioration in the strength of the control cables.